

L2 Interpreting scatter diagrams

Name: _____

Class: _____

Date: _____

Time: **77 minutes**

Marks: **60 marks**

Comments:

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

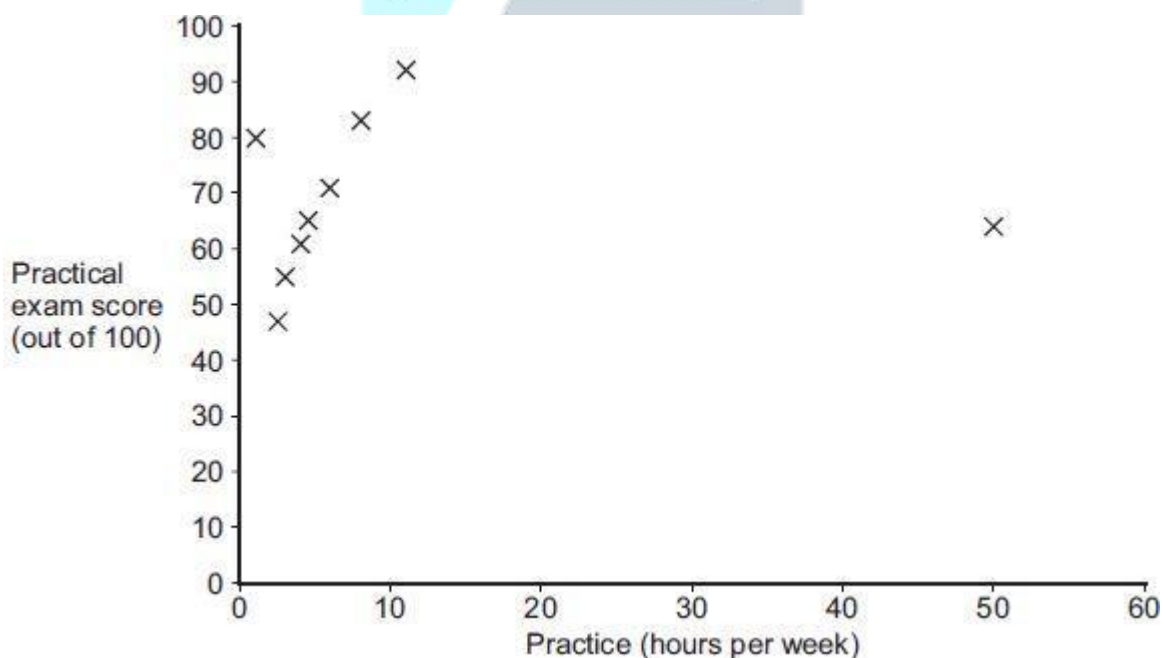
Q1.

Jennie is a piano teacher who teaches nine pupils.

She records how many hours per week they practice the piano along with their most recent practical exam score.

Student	Practice (hours per week)	Practical exam score (out of 100)
Donovan	50	64
Vazquez	6	71
Higgins	3	55
Begum	2.5	47
Collins	1	80
Coldbridge	4	61
Nedbalek	4.5	65
Carter	8	83
White	11	92

She plots a scatter diagram of this data, as shown below.



- (a) Identify two possible outliers by name, giving a possible explanation for the position on the scatter diagram of each outlier.

(4)

(b) Jennie discards the two outliers.

(i) Describe the correlation shown by the scatter diagram for the remaining points.

(1)

(ii) Interpret this correlation in the context of the question.

(1)

(Total 6 marks)

Q2.

Some information from the Large Data Set is given in Figures 1 and 2 below.

Figure 1

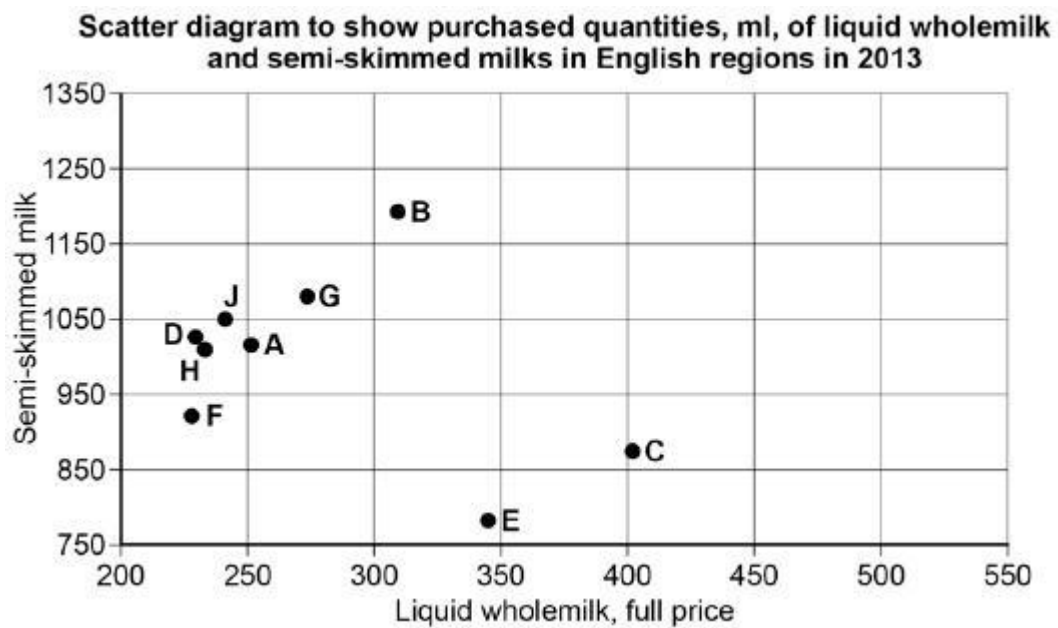
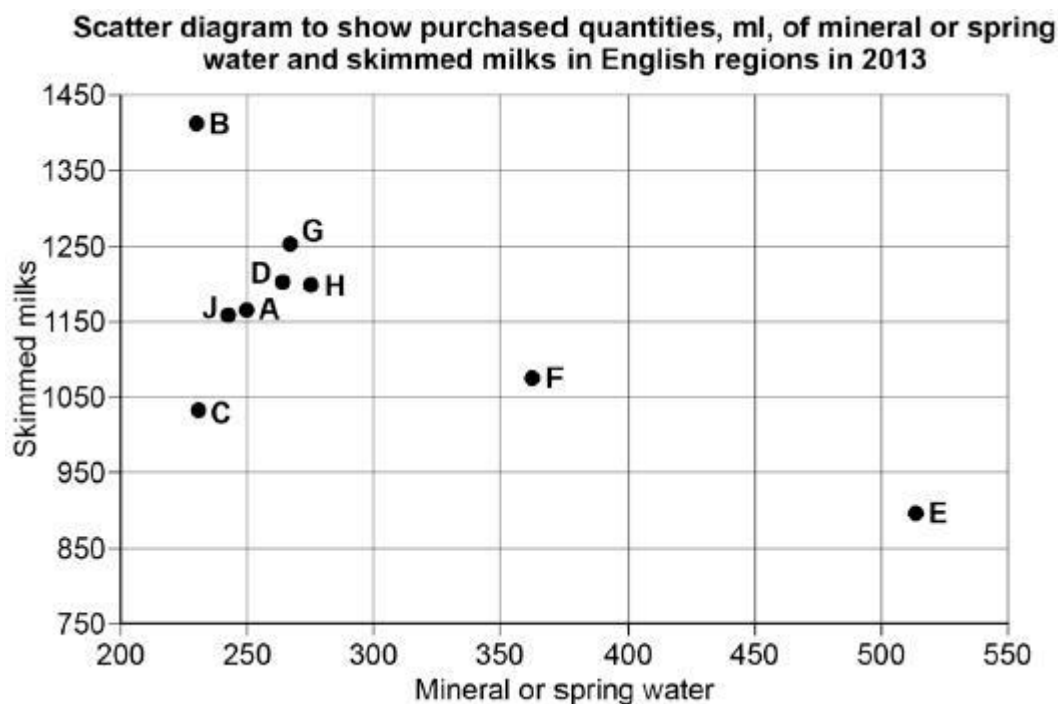


Figure 2



- (a) Give a reason why the recorded vertical data values are higher for each region in Figure 2 than in Figure 1

(1)

- (b) (i) Describe the correlation between 'Semi-skimmed milk' and 'Liquid wholemilk, full price'.

(2)

- (ii) Bilal claims that Figure 2 indicates that when people drink more mineral or spring water they tend to drink less skimmed milk.

Comment on Bilal's claim.

(2)

- (c) Suggest, with a reason, which region is indicated by the letter E. Use your knowledge of the Large Data Set to support your answer.

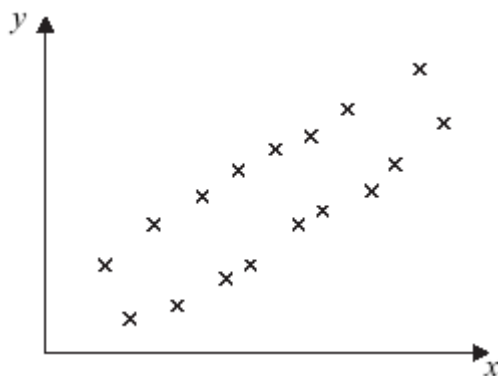
(2)

(Total 7 marks)

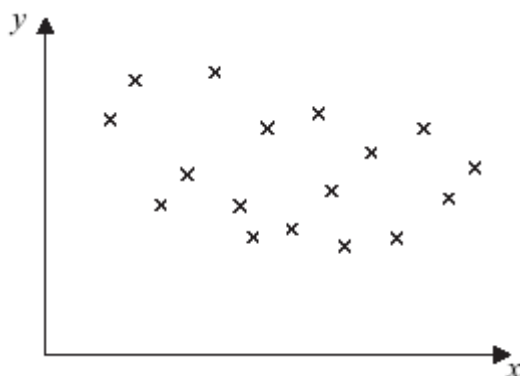
Q3.

Estimate, **without undertaking any calculations**, the value of the product moment correlation coefficient between the variables x and y for each of the two scatter diagrams.

- (a)



(b)



(Total 2 marks)

Q4.

Estimate, **without undertaking any calculations**, the value of the product moment correlation coefficient between the variables x and y in each of the two scatter diagrams.

(a)



(b)



(Total 4 marks)

Q5.

Monty, a young apprentice at a professional football club's academy for boys aged 11 to 18, is studying Statistics by a distance learning course. As part of his portfolio of work on correlation, he selects a sample of players from the club's academy and records their

ages, heights and weights.

Monty then calculates that the value of the product moment correlation coefficient for his results is:

- (a) -0.358 between age and height;
- (b) 1.14 between height and weight;
- (c) 0.469 between age and weight.

For **each** of Monty's calculations, state, with a reason, whether his result is probably correct, probably incorrect or definitely incorrect.

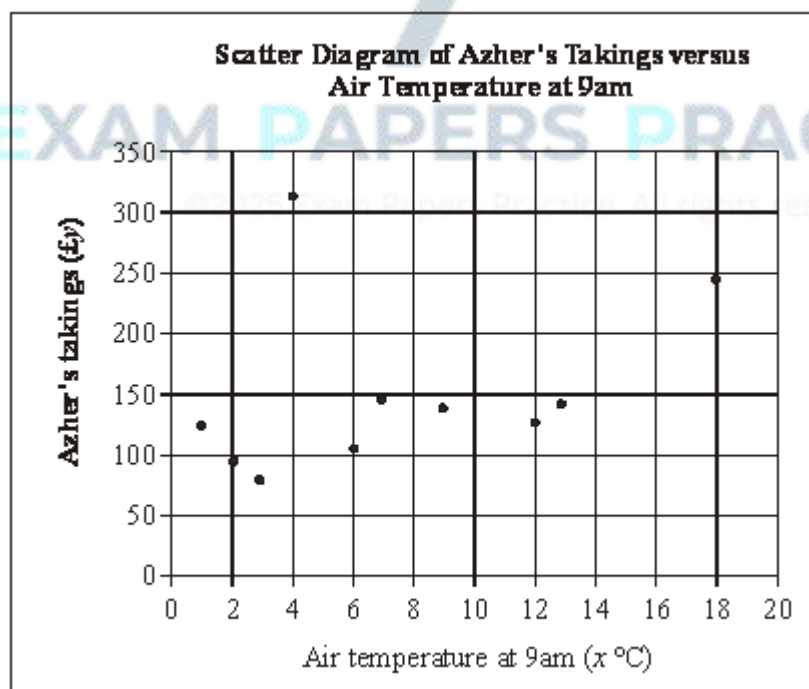
(Total 6 marks)

Q6.

Each Monday, Azher has a stall at a town's outdoor market. The table below shows, for each of a random sample of 10 Mondays during 2003, the air temperature, x °C, at 9 am and Azher's takings, £ y .

Monday	1	2	3	4	5	6	7	8	9	10
x	2	6	9	18	1	3	7	12	13	4
y	97	103	136	245	121	78	145	128	141	312

- (a) A scatter diagram of these data is shown below.



Give **two** distinct comments, in context, on what this diagram reveals.

(2)

- (b) One of the Mondays is found to be Easter Monday, the busiest Monday market of the year.

Identify which Monday this is most likely to be.

(1)

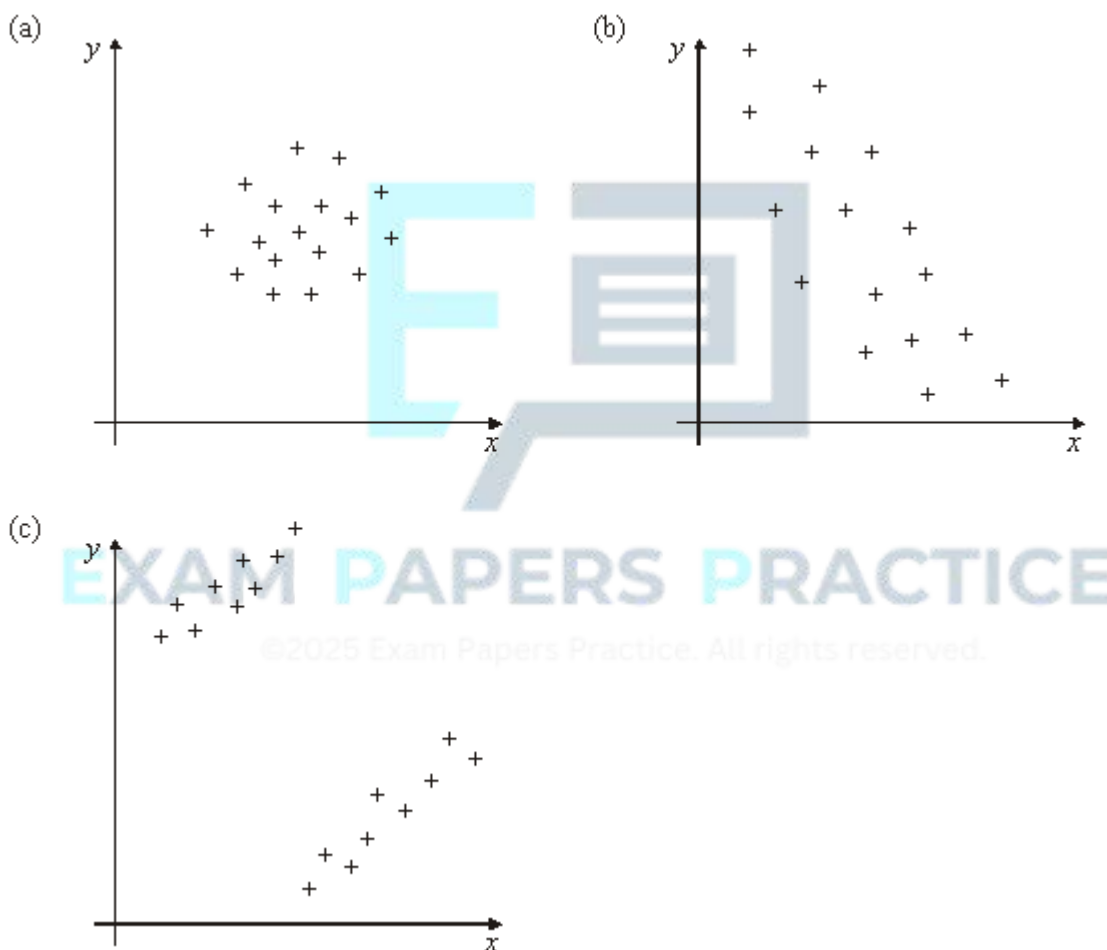
- (c) Name one other variable that would have been likely to affect Azher's takings at this town's outdoor market.

(1)

(Total 4 marks)

Q7.

Estimate, **without undertaking any calculation**, the value of the product moment correlation coefficient between the variables x and y in **each** of the scatter diagrams below.



(Total 5 marks)

Q8.

The following three statements are the results of Damien's calculations for a sample of towns in England.

Statement 1

The product moment correlation coefficient, between the percentage unemployed and the percentage of the population with qualifications beyond GCSE, is 0.65.

Statement 2

The product moment correlation coefficient, between the number of reported crimes of violence in 2003 and the number of chartered accountants, is 1.15.

Statement 3

The product moment correlation coefficient, between the number of reported crimes of violence in 2003 and the number of mathematics teachers, is 0.82.

Classify **each of the three** statements as either:

- A** plausible
- B** probably incorrect
- C** definitely incorrect

Give a reason for each of your answers.

(Total 6 marks)

Q9.

The scatter diagram below shows, for a sample of countries, average annual income per head, x , and the percentage of children in full-time education, y .

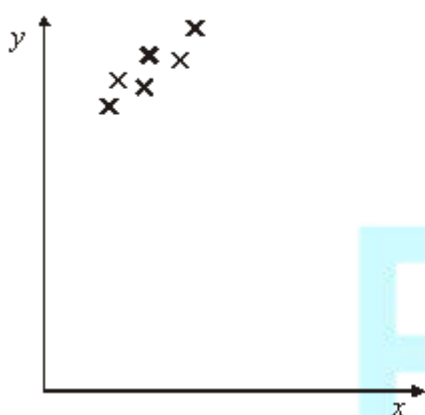
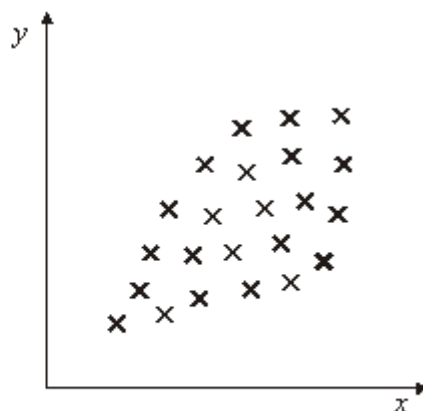
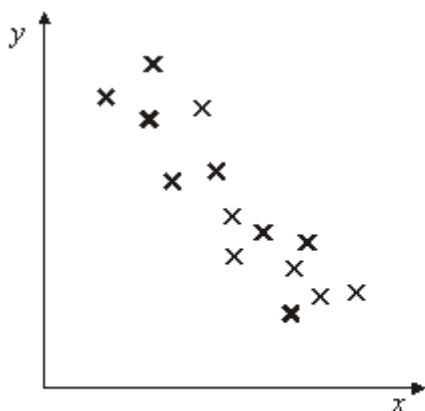


- (a) Describe the relationship between x and y .
- (b) Comment on the suggestion that the strength of the relationship should be measured by calculating the product moment correlation coefficient between x and y .

(2)
(Total 5 marks)

Q10.

Estimate, **without undertaking any calculations**, the values of the product moment correlation coefficient between the variables X and Y in **each** of the scatter diagrams below.



(Total 6 marks)

Q11.

An archaeologist has access to two scientific processes for determining the ages of ceramic fragments found in the excavation of ancient sites. The results for the ages, in hundreds of years, of ten fragments are as follows.

Fragment	A	B	C	D	E	F	G	H	I	J
Age process 1	152	123	105	145	107	128	115	143	135	128
Age process 2	155	121	107	144	105	126	112	144	139	129

- (a) (i) Plot a scatter diagram for the above data. (3)
- (ii) State what this indicates regarding the association between the ages given by the two processes. (2)
- (b) The value of the product moment correlation coefficient between the ages given by

process 1 and **process 2** is 0.992.

Stating the null and alternative hypotheses used, carry out a test at the 1% level of significance to investigate whether there is a direct association between the ages given by the two processes.

(4)

(Total 9 marks)



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.